To define TAE, it is assumed that the deviation of the measurement from the true value of the sample – called total error – consists of two parts, one systematic and the other random. In practical terms, the bias is considered as an estimate of the systematic error, and the standard deviation of precision combined with a coverage factor is an estimate of the random error dispersion. A more recent and practical definition of *TAE* given by International Council for Harmonization (ICH) is "the sum of the absolute value of the errors in accuracy (%) and precision (%)," where accuracy must be identified as trueness according to International Vocabulary of Metrology (VIM) definition [3].

From a mathematical point of view, the *TAE* is summarized by Eq. (6.1):

Total analytical error

$$TAE = \delta + k_{TAE} \times s_{TAE} \tag{6.1}$$

- δ the estimate of the lack of trueness in the form of a bias, which is supposed to be constant and systematic.
- s_{TAE} the estimate of precision (classically the intermediate precision standard deviation) assumed to cover the random part of measurement error.
- k_{TAE} the coverage factor accounting for the probability of coverage.

This approach has been considered attractive for its simplicity in clinical biology, as evidenced by guidelines in this domain of application [4]. But this attractiveness is more intellectual than practical, and *TAE* has not only advantages. The separate assessment of random and systematic errors is generally required by regulatory bodies and raises many practical difficulties. If the separation were possible, these two quantities would not have the same statistical property with respect to uncertainty.

If it seems understandable to regard the standard deviation of precision as related to a random variable (or a combination of random variables). It is more difficult to consider the bias of trueness as constant only because it is qualified as systematic. In practice, the bias, as it can be estimated in the analytical sciences, is not constant and varies from one replicate to another. These inconsistencies may explain the relatively limited use of *TAE* in other fields of analysis than clinical biology. It also explains why BIPM (and measurement's specialists) abandoned the error approach, in favor of the "uncertainty approach" as explained in Section 4.1.2.

6.2 General Procedure to Estimating MU

The principle of the GUM, a general procedure for estimating MU, is relatively simple, as we shall show [1]. It is called the GUM general procedure in the following pages. Several organizations have published documents, through application examples, to adapt them to the analytical sciences and the needs of laboratories [5]. Unfortunately, it became quickly evident that this adaptation is neither always flexible nor practical.

The general GUM procedure for estimating the MU is shown in Figure 6.1. It consists of four steps that will serve as a roadmap for the various examples of calculation explained in the following chapters. It is useful to introduce some notations.

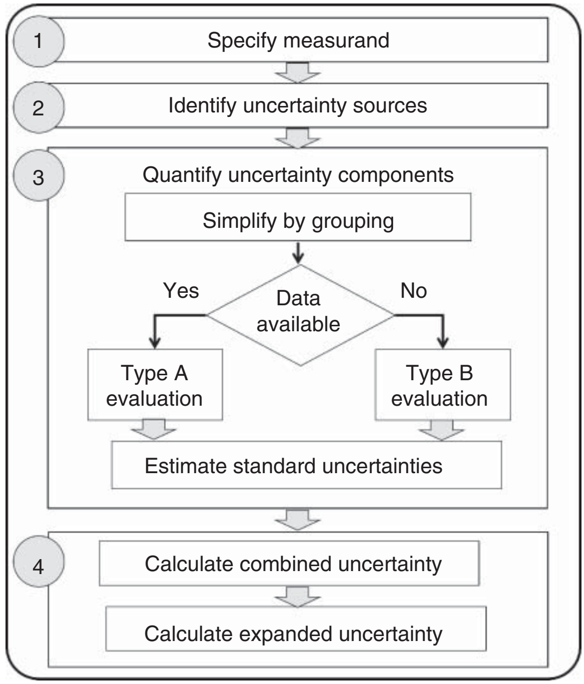

Figure 6.1 The 4-step GUM general procedure for measurement uncertainty (MU) estimation.

Let us keep Z to denote, as usual, the measured quantity of the analyte. MU can be expressed as four compatible parameters. Following the GUM conventions the following notations are used where k_{GUM} is the coverage factor:
Composed standard uncertainty

$$u_c(Z) \tag{6.2}$$

Expanded uncertainty

$$U(Z) = k_{GUM} \times u_c(Z) \tag{6.3}$$

Relative uncertainty

$$UR\%(Z) = \frac{U(Z)}{Z} \times 100 \tag{6.4}$$

Coverage interval $[I_L; I_U]$

$$\begin{aligned} &Z \pm k_{GUM} \times u_c(Z) \\ &Z \pm U(Z) \end{aligned} \tag{6.5}$$

The traditional values taken by the coverage factor k_{GUM} are defined in the GUM and presented below. Because analysts are inclined to use percentages in expressing MU (and many other parameters), the relative uncertainty is part of the list but not always used in other measuring domains. The L and U subscripts of the coverage interval are abbreviations of lower and upper, respectively.

6.3 Traceability at the International System of Units

When estimating the MU of a result, an important preliminary issue is the attachment of the measurand to the International System of Units (SI), i.e. how the traceability of measurement values to official standards is ensured. For chemistry and biology, as stated, the SI unit of measure is the mole, whose symbol has been mol since 1970. Earlier, in the middle of the nineteenth century, the molar mass of the isotope 12 of carbon, denoted $M(^{12}C)$ was fixed at exactly 12.00000 g/mol. From there, by applying different methods of measurement, it was acknowledged by convention that a mole contained about 6×10^{23} elementary entities. This number was called the Avogadro number at the beginning of the twentieth century. It is noted N_A and initially came with a limited number of significant figures and an uncertainty that regularly decreased as measurement techniques improved. Until 2018, this reference to ^{12}C was the accepted definition of the mole. But at the 26th meeting of the General Congress of Weights and Measures (CGPM), it was decided to completely overhaul the SI system by defining seven constants from which all units of measurement were to be derived. The crucial point is that these constants are considered to have no MU. From then on, the Avogadro's *number* became Avogadro's *constant* which today is exactly:

$$N_A = 6.02214076 \times 10^{23}$$

Consequently, the molar mass of carbon 12 is no longer constant but has a standard uncertainty:

$$M(^{12}C) = 12.01074 \pm 0.00047\ \text{g/mol}$$

The molar mass of a molecule is determined in relation to the molar masses of each constitutive atom. For example, the empirical formula of theophylline is $C_7H_8N_4O_2$. The following table shows how to calculate its molar mass and the associated MU.

Element	Atomic weight M	$u(M_i)$	n_i	$n_i \times M_i$	$u^2(M_i)$	$n_i^2 \times u^2(M_i)$
C	12.0106	0.0006	7	84.0742	$3.60\ 10^{-7}$	$1.764\ 10^{-5}$
N	14.00686	0.00025	4	56.02744	$6.25\ 10^{-8}$	$1.000\ 10^{-6}$
O	15.9994	0.00021	2	31.9988	$4.41\ 10^{-8}$	$1.764\ 10^{-7}$
H	1.00798	0.00008	8	8.06384	$6.40\ 10^{-9}$	$4.096\ 10^{-7}$
Total (g/mol)				180.1643		$19.226\ 10^{-6}$

The atomic weights and their standard uncertainties used for the calculation come from the Internet site of the Commission on Isotopic Abundances and Atomic Weights (CIAAW) managed by International Union for Pure and Applied Chemistry (IUPAC) [6]. They are normalized atomic weights established by considering an average isotopic composition. This explains the difference between the carbon atomic weight used for calculation and the exact value given above because normalized carbon is a mixture of several isotopes.

The final molar mass of the molecule can thus be modified when it is marked. Since the total mass is an additive polynomial, the combined uncertainty is obtained by going through the standard variances of each constituent atom as explained in Section 6.7.1: this rationale is called the variance propagation law. If n_i the number of ith atom in a given molecule, finally:

Total molar mass g/mol	$\sum n_i \times M_i$	180.1643
Standard variance u^2	$\sum n_i^2 \times u^2(M_i)$	$19.226\ 10^{-6}$
Expanded uncertainty U g/mol	$2 \times \sqrt{u^2}$	0.0088
Relative uncertainty $UR\%$		0.0049%

In the case of a calibrator solution obtained by weighting, the contribution of the molecule mass uncertainty to the overall MU is generally exceedingly modest compared to the other input quantities involved. For the molecular mass of theophylline, the relative uncertainty is only 0.0049%. But there are some special examples, such as gas analysis, where the contribution of molecular mass MU can be more important. However, the wider use of the mole as a unit in analytical science has two drawbacks:

- Its creation is recent, whereas the analytical sciences have a much older history and tradition to quantify a result.
- Unlike the meter or the kilogram when initially created, no concrete standards are available allowing an easy attachment to the mole. To have such standards, thousands or even millions of items would be needed, one for each chemical molecule. In a way, certified reference materials (CRM) play this role, but for a limited number of applications, as explained in Section 6.4.

Today most standards are dematerialized. For instance, the meter is no longer defined in reference to the prototype bar of platinum but as “the length of the path travelled by light in a vacuum in 1/299792458 of a second.”

When the mole is defined in a comparable manner, many problems will be solved for analysts. However, the mole is not that often used in the laboratories, and it is frequent to express a concentration using other units, such as a mass (kg), a relative mass or a mass fraction (kg/kg or kg/l) which allows a connection to another more convenient SI unit, the kg. The downside of a relative concentration is that it is a dimensionless quantity. A widespread practice then – although discouraged by the BIPM – is to use a fraction to express a concentration, such as percent (or 10^{-2}), ppm (part per million or 10^{-6}) or ppb (part per billion or 10^{-9}).

In solution chemistry, when it is necessary to consider the ionized form of a molecule, the Equivalent, noted Eq, is applied. It is a unit which integrates the electric charge and the mole, but which is not part of the SI. Thus, for the ions which carry a charge +1 or −1, like Na^+, HCO_3^- or Cl^-, 1 mol = 1 Eq. For the ions of valence +2 or −2, as Ca^{2+}, 1 mol = 2 Eq. and so on for the other valence values.

From these units, various secondary units are derived, as well as fractions of units, such as the nanogram (ng), the millimole (mmol) or the milliequivalent (mEq). Finally, in some fields, such as biomedical analysis or microbiology, many analytes are poorly defined in terms of molecular structure, such as macromolecules, or mixtures of molecules. The solution adopted is to use the International Unit (IU) system, which is misnamed because it has nothing to do with the SI units. The definition of the IU differs from one substance to another.

It is the World Health Organization (WHO) Biological Standardization Committee that defines the IU of a substance based on the experimental measurement of its biological effects. For example, a 1000 IU tablet of oxytetracycline contains 1.149 mg of this antibiotic or 1.136 mg in its bi-hydrate form. A 1000 IU/ml preparation of vitamin D (a mixture of molecules with a vitamin D effect) contains 0.025 mg/ml.

6.4 Stage 1. Specify the Measurand

The four stages of the GUM general procedure are described in Sections 6.4–6.8.

Stage 1 starts with the specification of the measurand which is the "quantity intended to be measured" as explained in Chapter 1. In the analytical sciences, the term analyte is usually preferred to measurand. It is a chemical or biochemical species sought by the analyst and present in a matrix at known or unknown quantity. A measurement value of this quantity is generally the combination of:

- An operating procedure (i.e. a sequence of operations) that allows to prepare the sample by extracting the analyte from the rest of the matrix. It is done by playing on selectivity so as to obtain the analyte in such a form that it can be introduced into a measuring device without damaging it, or even to concentrate it to obtain a higher and detectable instrumental signal.
- And several input quantities, which are according to VIM "quantity that must be measured, or a quantity, the value of which can be otherwise obtained, to calculate a measured value of a measurand." They mainly represent an instrumental signal, a weight, a volume, or adjustment and correction coefficients that enter in the measurement model.

Situations are extremely diverse between non-destructive methods, indirect methods with calibration, primary methods, etc. The identification of the measurand consists precisely in describing and considering all these elements. As a first estimate of MU, the preanalytical operations of sampling will be neglected but fully addressed in a specific Section 8.3. As generally proposed in the guides designed for analysts, only analytical input quantities are taken into consideration.

The next chapter explains how to account for all the components of the MU in the case of a measurement obtained in a chemical or biological laboratory. With this simplified approach, the expression of a measurement is essentially based on the mathematical formula used to calculate the result. This is called the Measurement Model and consists of the mathematical relationship between the analyte and the various input quantities, such as reagent purities, the coefficients of the